Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS5IOT05]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

SENSOR TECHNOLOGIES

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	A sensor has to detect temperature and display in Celsius, Fahrenheit and Kelvin, considering the above scenario Discus the functional elements of measuring system	**06**
	b.	Define transducer. List classification of transducer. Explain Active Transducer With the help of diagram.	**08**
2.	**a.**	Illustrate RTD. Why Platinum is the most commonly used metal for RTD elements? What are the factors on which it depends?	**08**
	b.	With the help of diagram explain the principle of operation, construction details and Characteristics of LDR.	**06**
3.	**a.**	Illustrate capacitive touch screen sensor and its basic elements.	**08**
	b.	Explain construction and Working details of LVDT.	**06**
		OR	
4.	**a.**	Discuss clap switch circuit with the help of capacitor microphone	**07**
	b.	Analyze the relation between capacitance, area and distance between its two plates.	**07**
5.	**a.**	Explain working principle of flex sensor with examples	**04**
	b.	Write a short on 1. Dust Sensor 2.Barometer	**10**
		OR	
6.	**a.**	Explain digital fiber optical transducer and list its different types.	**07**
	b.	"In an online attendance system based on smart tag faculties are taking attendance" Considering the above application explain following. **i.** Working principle of RFID tag. **ii.** How faculty can take attendance. **iii.** Disadvantage of RFID tag.	**07**

7. a. List the sensors which checks water quality and explain principle of operation of any one.

b. Draw and explain block diagram of weather monitoring system.

OR

8. a. Draw a neat diagram to demonstrate home automation system with the help of sensors and actuators. **07**

b. Design a smart vehicular monitoring system. Show sensors and actuators used. **07**